

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a938ba31ce83abe60.jsonl`  
- SHA-256: `ff440da324ce66e889a49e04f605435a7ae5abfbf41213f44f85c08da78c83a5`  
- Source modified: `2026-05-31T04:24:44+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:23:18.724Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"On olmOCR-Bench the model scores 82.3% overall with 84.3% on Tables and 84.3% on Multi-column layouts ? directly relevant to dense, multi-column bank-statement tables."

****Source:**** <https://huggingface.co/allenai/olmOCR-2-7B-1025> (primary)

****Supporting quote:**** "Tables | 84.3% ... Multi-column | 84.3% ... Overall | 82.3 ± 1.1%"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:23:21.255Z)

I'll verify this claim about olmOCR-2-7B-1025 benchmark scores. Let me start by loading the web tools and fetching the primary source.

3. assistant (2026-05-31T04:23:24.402Z)

I'll fetch the primary source to verify the exact benchmark numbers claimed.

4. user (2026-05-31T04:23:30.200Z)

olmOCR-2-7B-1025 Benchmark Scores and License

olmOCR-Bench Scores

Based on the model card, here are the exact benchmark scores when used with the olmOCR toolkit (v0.4.0):

Category	Score
----- -----	
Overall	82.3 ± 1.1
ArXiv	82.9
Old Scans Math	82.1
Tables	84.3
Old Scans Headers and Footers	48.3
Multi-column	84.3
Long tiny text	81.4
Base	99.7

Confidence Interval: ± 1.1 (reported for Overall score)

License

Apache 2.0 - The model weights are released under the Apache 2.0 license and are intended for research and educational use in accordance with Ai2's [Responsible Use Guidelines](https://allenai.org/responsible-use).

Note: The model card also mentions that the FP8 quantized version ([olmOCR-2-7B-1025-FP8](https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8)) achieves nearly identical scores (82.4 ± 1.1 overall) and is recommended for practical use except for further fine-tuning.

5. assistant (2026-05-31T04:23:32.550Z)

The primary source confirms the exact numbers in the claim. Let me verify these aren't cherry-picked or contradicted by searching for independent coverage, and confirm the dataset/methodology context.

6. user (2026-05-31T04:23:40.263Z)

Web search results for query: "olmOCR-2-7B-1025 olmOCR-Bench 82.3 benchmark tables multi-column"

Links: [{"title": "OlmOCR-Bench: OCR Evaluation Benchmark", "url": "https://www.emergentmind.com/topics/olmocr-bench"}, {"title": "olmOCR 2: Unit test rewards for document OCR | Ai2", "url": "https://allenai.org/blog/olmocr-2"}, {"title": "[2510.19817] olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://arxiv.org/abs/2510.19817"}, {"title": "olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://arxiv.org/pdf/2510.19817"}, {"title": "richardyoung/olmOCR-2-7B-1025-MLX-6bit · Hugging Face", "url": "https://huggingface.co/richardyoung/olmOCR-2-7B-1025-MLX-6bit"}, {"title": "(PDF) olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://www.researchgate.net/publication/396790587_olmOCR_2_Unit_Test_Rewards_for_Document_OCR"}, {"title": "olmOCR 2 Unit Test Rewards for Document OCR", "url": "https://arxiv.org/html/2510.19817v1"}, {"title": "allenai/olmOCR-2-7B-1025-FP8 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8"}, {"title": "olmOCR-2-7B-1025 OCR Model", "url": "https://www.emergentmind.com/topics/olmocr-2-7b-1025"}, {"title": "allenai/olmOCR-2-7B-1025 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025"}]

Based on the search results, here's what I found about olmOCR-2-7B-1025 and olmOCR-Bench:

Overview

olmOCR 2 (olmOCR-2-7B-1025) achieves state-of-the-art performance for real-world OCR of English-language digitized print documents. It's a specialized 7B vision language model (VLM) trained using reinforcement learning with verifiable rewards (RLVR), where rewards are a diverse set of binary unit tests.

Benchmark Performance

On olmOCR-Bench, olmOCR 2 achieves 82.4 points—a nearly +4-point improvement over the previous release. (Note: The search results show 82.4 rather than 82.3, which may reflect the most recent scoring.)

Improvements on Multi-Column and Table Parsing

olmOCR-2-7B-1025 achieves improvements of +22.6 (Tables) and +16.1 (Multi-column) percentage points. The largest improvements are in math formula conversion, table parsing, and multi-column layouts compared to previous versions.

About olmOCR-Bench

OlmOCR-Bench serves as the primary evaluation suite for state-of-the-art end-to-end OCR and vision-LLMs, including olmOCR-2-7B-1025. OlmOCR-Bench eschews traditional error rates such as CER and WER in favor of strict, deterministic, task-specific unit tests.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-05-31T04:23:43.005Z)

Web search results for query: "olmOCR-Bench leaderboard 2025 tables multi-column scores comparison"

Links: [{"title": "OlmOCR-Bench: OCR Evaluation Benchmark", "url": "https://www.emergentmind.com/topics/olmocr-bench"}, {"title": "7 Best Open-Source OCR Models 2025: Benchmarks & Cost Comparison | E2E Networks", "url": "https://www.e2enetworks.com/blog/complete-guide-open-source-ocr-models-2025"}, {"title": "olmOCR-Bench Leaderboard | OCR Benchmark | CodeSOTA | CodeSOTA", "url": "https://www.codesota.com/ocr/benchmark/olmocr-bench"}, {"title": "allenai/olmOCR-bench · Datasets at Hugging Face", "url": "https://huggingface.co/datasets/allenai/olmOCR-bench"}, {"title": "olmOCR Benchmark 2026: 1 model averages | BenchLM.ai", "url": "https://benchlm.ai/benchmarks/olmOcr"}, {"title": "GitHub - allenai/olmocr: Toolkit for linearizing PDFs for LLM datasets/training", "url": "https://github.com/allenai/olmocr"}, {"title": "OlmOCR-Bench Review: Insights and Pitfalls | LlamaIndex", "url": "https://www.llamaindex.ai/blog/olmocr-bench-review-insights-and-pitfalls-on-an-ocr-benchmark"}]

Based on the web search results, here's what I found about the olmOCR-Bench leaderboard for 2025 with scores across different task categories:

Leaderboard Overview

The olmOCR-Bench leaderboard includes 55 models, 22 metrics, and was last updated on April 20, 2026. The leaderboard shows open-source models scoring 75-83% on challenging document parsing tasks.

Top Models and Overall Scores

According to the Hugging Face leaderboard, top models include: Infinity-Parser2-Pro (87.6), Chandra-OCR-2 (85.9), dots.mocr (83.9), Surya-OCR-2 (83.3), and LightOnOCR-2-1B (83.2).

Multi-Column Performance

OlmOCR-2-7B-1025 achieved improvements including +22.6 percentage points on Tables and +16.1 percentage points on Multi-column.

Task Categories

The benchmark includes 9 metrics across categories including: base, headers-footers, long-tiny-text, multi-column, arxiv, tables, old-scans-math, and pass rate. Table parsing involves cell-wise placement and content verification, while multi-column layout validates natural reading order across multiple columns.

The actual detailed leaderboard tables with specific scores for each category are available on the CodeSOTA platform and Hugging Face dataset pages linked in the search results.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-05-31T04:23:52.270Z)

The searches confirm and contextualize the claim well. Let me note the key findings:

1. **Primary source (HF model card) confirms the exact numbers**: Overall 82.3 ± 1.1 , Tables 84.3%, Multi-column 84.3%. These match the claim verbatim.

2. **Independent corroboration**: The Ai2 blog and arXiv paper (2510.19817) confirm the model exists and the improvements on Tables (+22.6 pp) and Multi-column (+16.1 pp) ? the two categories highlighted in the claim are genuinely the model's strongest improvement areas, so they aren't arbitrary cherry-picks.

3. **Minor discrepancy worth noting**: The Ai2 blog cites "82.4" overall, and the FP8 variant scores 82.4 ± 1.1 . The 82.3 figure is the non-quantized base model. Within the ± 1.1 CI these are statistically indistinguishable. The claim uses 82.3, which is the correct figure for the exact model cited (olmOCR-2-7B-1025, non-FP8). This is accurate, not an error.

Now let me consider potential refutation angles. The claim's research-relevance assertion ("directly relevant to dense, multi-column bank-statement tables") deserves scrutiny ? olmOCR-Bench is built on **English-language digitized print documents** (arXiv papers, old scans), not financial statements. Let me check the LlamaIndex critique for benchmark caveats and whether the benchmark is contested.

9. user (2026-05-31T04:23:59.631Z)

OlmOCR-Bench Criticisms: Tables and Multi-Column Limitations

The article identifies several significant methodological concerns about the benchmark's reliability for real-world document parsing:

Table Score Reliability Issues

The author argues that "The binary unit tests are still quite coarse" for tables specifically. Rather than comprehensively evaluating table quality, the tests only check "narrow adjacency relationships ? essentially 'is cell X above cell Y?'" This means models can pass most tests while producing "globally incorrect or incomplete outputs." The benchmark misses critical real-world complexities like "merged cells," "spanning," and overall "table structure" completeness.

Additionally, the benchmark has a fundamental selection bias: the creators "throw away any data that Gemini 2.0 Flash cannot parse," meaning the dataset only includes tables that frontier models could already handle. This artificially inflates apparent performance.

Methodological Concerns

The article questions whether the benchmark reflects actual business document needs. The dataset is "~56% academic," focused on "arXiv math, old textbooks, multi-column journal pages" rather than invoices, forms, or financial statements. This skews the multi-column category toward academic layouts, not diverse real-world variations.

The author doesn't directly dispute olmOCR's reported scores but suggests they're misleading?high scores on coarse-grained tests don't guarantee utility for practical document

workflows.

10. assistant (2026-05-31T04:24:11.562Z)

This is a crucial finding for the adversarial review. The LlamaIndex critique substantively qualifies the claim's relevance assertion. Let me weigh the two halves of the claim separately.

Part A ? the raw numbers ("scores 82.3% overall with 84.3% on Tables and 84.3% on Multi-column"):

- Fully supported by the primary source (HF model card), verbatim.
- Corroborated by the Ai2 blog and arXiv paper.
- The numbers are accurate and current (Oct 2025 release, 1025 = release date).
- This half is NOT refutable.

Part B ? the relevance assertion ("directly relevant to dense, multi-column bank-statement tables"):

- This is the appended interpretive claim, and it is materially overstated. The LlamaIndex review documents that:
 1. olmOCR-Bench's "Tables" unit tests are **coarse** ? they only check pairwise cell adjacency ("is cell X above cell Y?"), so a model can pass while producing globally incorrect/incomplete tables. This is exactly the failure mode that matters for bank statements (dropped rows, transposed digits).
 2. The dataset is **~56% academic** (arXiv math, old textbooks, multi-column journals), with the multi-column category skewed toward academic layouts ? **not** invoices, forms, or financial statements.
 3. **Selection bias**: data Gemini 2.0 Flash couldn't parse was discarded, inflating apparent performance.
- The official scope is explicitly "English-language digitized print documents," not dense Indian bank statements.

So the benchmark scores do **not** straightforwardly translate to "directly relevant to dense, multi-column bank-statement tables." The 84.3% Tables score in particular measures something much weaker than full table-structure correctness on financial data.

The claim as a whole makes a verbatim-accurate factual statement and then attaches an interpretive leap. The numerical core is correct; the "directly relevant to bank-statement tables" framing is an overreach contradicted by credible critique of what the benchmark actually measures. Given this is voter 2/3 and the instruction to be skeptical of overreach, the relevance claim is the load-bearing inference for the research question, and it does not hold up cleanly.

Let me make one more check on the arXiv paper to confirm the benchmark's stated domain scope, to ensure I'm not over-relying on a single critic.

11. user (2026-05-31T04:24:16.771Z)

olmOCR-Bench Document Types and Evaluation Methodology

Based on the provided content, the paper does not specify which document types olmOCR-Bench evaluates. The abstract mentions olmOCR 2 converts "digitized print documents, like PDFs, into clean, naturally ordered plain text" and notes improvements in "math formula conversion, table parsing, and multi-column layouts," but it does not detail whether the benchmark includes financial documents, invoices, forms, or focuses specifically on academic materials.

Regarding table and multi-column evaluation methods, the abstract provides no description of the actual unit tests or evaluation methodology. It only states that the model showed "the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions."

To obtain specifics about document types included in olmOCR-Bench and detailed table/multi-column evaluation criteria, you would need to consult the full paper itself, which is available via the PDF link provided on the arXiv page.

12. assistant (2026-05-31T04:24:27.758Z)

The abstract doesn't go deep enough, but the official Ai2 description ("digitized print documents") and the LlamaIndex review (which examined the actual dataset composition and unit-test design) are sufficient corroboration. The benchmark is print-document-focused (~56% academic), and the table tests check coarse pairwise adjacency rather than full structural correctness.

I have enough to render a verdict. Let me summarize my assessment.

Verdict: REFUTED (with nuance).

The claim has two parts joined by an em-dash:

1. **The numbers** ? "82.3% overall, 84.3% Tables, 84.3% Multi-column." These are **verbatim accurate** per the primary HF model card and corroborated by the Ai2 blog/arXiv paper. This half is sound.
2. **The relevance inference** ? "directly relevant to dense, multi-column bank-statement tables." This is an **overreach** that credible evidence contradicts:
 - olmOCR-Bench is built on **English-language digitized print documents (~56% academic)**: arXiv math, textbooks, journal pages), explicitly NOT financial statements/invoices/forms.
 - The "Tables" unit tests are **coarse** ? they verify only pairwise cell adjacency ("is cell X above cell Y?"), so a model can score 84.3% while producing globally incomplete/incorrect tables. This does not measure the full table-structure fidelity (no dropped/transposed rows) that bank-statement parsing requires.
 - **Selection bias**: examples Gemini 2.0 Flash couldn't parse were discarded, inflating apparent scores.
 - The multi-column category is skewed toward academic layouts, not dense semi-ruled financial tables.

Per the rubric, refuted=true applies when a claim is an overreach beyond what the source supports. The load-bearing inference for the research question ("directly relevant to bank-statement tables") is the overreach, even though the numbers themselves are correct. I'll flag this as the specific defect, with medium confidence (the numerical core is unimpeachable; only the interpretive leap fails).

13. user (2026-05-31T04:24:36.601Z)

Structured output provided successfully

14. assistant (2026-05-31T04:24:44.564Z)

Verdict delivered: **refuted=true** (medium confidence).

Summary of my adversarial review:

- **The numerical claim is verbatim accurate** ? the primary HF model card reports exactly 82.3% overall (± 1.1), 84.3% Tables, and 84.3% Multi-column, with Apache-2.0 weights. Corroborated by Ai2's blog and arXiv 2510.19817. The numbers themselves survive scrutiny.
- **The refutation targets the load-bearing inference** ? "directly relevant to dense, multi-column bank-statement tables." Three independent problems undercut it:
 1. olmOCR-Bench is ~56% academic print documents (arXiv math, textbooks, journals), explicitly not financial statements/invoices/forms.
 2. The "Tables" unit tests check only coarse pairwise cell adjacency, so 84.3% does not certify the dropped-row/transposed-digit fidelity bank-statement parsing requires (per LlamaIndex's benchmark critique).
 3. Selection bias ? examples Gemini 2.0 Flash couldn't parse were discarded, inflating scores.

The claim pairs a correct fact with an overstated relevance assertion to the research question. Medium confidence because the numerical core is unimpeachable; only the interpretive leap fails ? a downstream report should cite the scores but explicitly caveat that olmOCR-Bench is print/academic-focused and its table metric is coarse, so the model still needs muneem's own balance-reconciliation verification on real Indian bank statements before being trusted.